



SAFETY Department
SIMHADRI

Note No :1



Ref:SMTTP/SAFETY/2023-24/NOTE/635021

Date:

07/11/2023(17:28:58)

Subject:Approval of LMI on IMPLEMENTATION OF SAFETY RULES - ELECTRICAL & MECHANICAL

Subject: Approval of LMI on "Implementation of Safety Rules - Electrical & Mechanical" - reg.

The LMI on Implementation of Safety Rules - Electrical and Mechanical, Ref no. LMI - OPS-SAFT-002 Rev 1- Dated: 07.11.2023 is reviewed and changes incorporated in the LMI as follows:

In Clause 5.1.1, paragraph 2 is edited as per requirement of procedural compliance check

This supercedes LMI - OPS-SAFT-002 Rev 0, Dated: 03.09.2021.

Put up for approval of Competent authority.

Submitted by:

पुली सदाशिव रेड्डी - उप महाप्रबंधक

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On:07/11/2023(17:28:58) **Ref:**SMTTP/SAFETY/2023-24/NOTE/635021

Note No:2

Checked and put up for approval of Competent Authority

Submitted by:

तुषार कानूनगो - अपर महाप्रबंधक

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Note No:3

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Note No:4

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On:08/11/2023(09:21:39) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:5

Please review.

Submitted by:

राजेश चावला - अपर महाप्रबंधक

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On:08/11/2023(11:06:49) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:6

Reviewed. Found in order.

Submitted by:

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On:09/11/2023(09:16:58) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:7

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Note No:8

Submitted by:

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Note No:9

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Note No:10

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Note No:11

Names mentioned in page 2 to be corrected.

Submitted by:

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On:10/11/2023(10:17:52) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:12

plz refer the enclosure

LMI - OPS - SAFT - 002 Rev 1- IMPLEMENTATION OF SAFETY RULES ELECTRICAL AND
MECHANICAL- 07.11.2023.doc.

Submitted by:

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On:10/11/2023(15:30:02) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:13

Submitted for further approval.

Submitted by:

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On:10/11/2023(15:35:22) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:14

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Note No:15

Submitted by:

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On:14/11/2023(08:59:57) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:16

Reviewed.

May please be approved.

Submitted by:

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On:23/11/2023(10:59:14) **Ref:**SMTPP/SAFETY/2023-24/NOTE/635021

Note No:17

Pl review.

Submitted by:

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On:23/11/2023(17:35:11) **Ref:**SMTPP/SAFETY/2023-24/NOTE/635021

Note No:18

Reviwed & Found in order Pl.

Submitted by:

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On:27/11/2023(12:53:44) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:19

Submitted by:

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On:27/11/2023(15:48:42) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:20

Reviewed and submitted for approval

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

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On:28/11/2023(16:23:42) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:21

May please review

Submitted by:

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On:30/11/2023(09:33:54) **Ref:**SMTPP/SAFETY/2023-24/NOTE/635021

Note No:22

Please review

Submitted by:

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On:30/11/2023(10:12:42) **Ref:**SMTPP/SAFETY/2023-24/NOTE/635021

Note No:23

Reviewed and found in order.

A minor correction in the document is required wherein it is written

"Compliance checks & audits of the implementation of SR-EM will be of **three types**"

but four nos of points are mentioned.

May please be corrected and forwarded for further approval.

Submitted by:

रश्मि रंजन सतपथी - वरिष्ठ प्रबंधक

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On:03/12/2023(13:07:16) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:24

Reviewed and forwarded for further approval.

Submitted by:

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On:03/12/2023(14:34:13) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:25

Submitted by:

शंभू कुमार सुमन - अपर महाप्रबंधक

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On:04/12/2023(10:07:57) **Ref:**SMTPP/SAFETY/2023-24/NOTE/635021

Note No:26

Reviewed and forwarded for further approval.

Submitted by:

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On:04/12/2023(17:29:24) **Ref:**SMTPP/SAFETY/2023-24/NOTE/635021

Note No:27

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सोमनाथ राउत - अपर महाप्रबंधक

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On:04/12/2023(17:54:08) **Ref:**SMTPP/SAFETY/2023-24/NOTE/635021

Note No:28

Only the first page of the SR-EM is attached at the end of the LMI. The OD has the full version. May like to add the full document of SR-EM to make the LMI complete in itself.

Submitted by:

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On:04/12/2023(18:28:50) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:29

SR-EM is enclosed as advised

Submitted by:

पुली सदाशिव रेड्डी - उप महाप्रबंधक

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On:06/12/2023(15:48:38) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:30

Submitted by:

कासा वेंकटेश्वर प्रसाद - अपर महाप्रबंधक

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On:06/12/2023(17:40:00) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:31

Submitted by:

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On:06/12/2023(18:18:53) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:32

Noted

Submitted by:

देबजीत कुमार कर - अपर महाप्रबंधक

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Note No:33

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जगनमोहन राव कोनेटी - महाप्रबंधक

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On:12/12/2023(12:42:20) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:34

Pl review

Submitted by:

टी.आर. राजगोपालन् - महाप्रबंधक

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On:13/12/2023(16:25:55) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:35

Reviewed and is forwarded for approval.

Submitted by:

मलय कांति जाना - अपर महाप्रबंधक

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On:14/12/2023(17:51:49) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:36

The SR-EM (Safety Rules, Electrical and Mechanical) shall be an integral part of this LMI and the same is given in the annexures as a ,pdf document which shall form an integral part of this LMI. CA may please accord approval for the revision of this LMI.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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On:19/12/2023(10:20:51) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021

Note No:37

Approved & Submitted by:

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On:23/12/2023(19:10:11) **Ref:**SMTTPP/SAFETY/2023-24/NOTE/635021



SIMHADRI

NTPC LIMITED
SIMHADRI SUPER THERMAL POWER STATION

Local Management Instruction – LMI-OPS-SAFT-002

(COS-ISO-00-OD/OPS/SAFT/002 -Rev. No. 1, Dated: 01.11.2023)

Revision: 01

Dt: 01.11.2023

IMPLEMENTATION OF SAFETY RULES
(ELECTRICAL & MECHANICAL)

Approved for Implementation by.....

HEAD OF PROJECT (Simhadri)

Date: .11.2023

TITLE DOCUMENT

TITLE OF LMI:		Implementation of Safety Rules (Electrical & Mechanical) at Simhadri Super Thermal Power Station.					
LMI No:		LMI-OPS-SAFT-002					
ISSUE NO	REVISION NO	DATE	CHANGES DONE	PREPARED BY	CHECKED BY	REVIEWED BY	APPROVED BY
00	00	.09.21	All clauses are new as the document is being issued for the first time.	P S S REDDY	ALL HODs	GM (O&M)	HOP
00	01	.11.23	In Clause 5.1.1, paragraph 2 is edited as per requirement of procedural compliance check	P S S REDDY	ALL HODs	GM (O&M)	HOP

N T P C L I M I T E D

**SIMHADRI SUPER THERMAL POWER STATION
LMI APPROVAL DOCUMENT**

Title of LMI:	Implementation of Safety Rules (Electrical & Mechanical) at Simhadri Super Thermal Power Station	
LMI No.:	LMI-OPS-SAFT-002	
Issue No. :0	Revision: 01	Date: 07.11.2023

	Name Sri.	Designation	Signature
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Checked by	Tushar Kanungo	AGM (Safety)	
	Uma Shankar Verma	AGM (EEMG)	
	Rajesh Chawla	AGM (MTP)	
	Dr. Indrajeet Jena	Senior Specialist	
	Nilanjan Mondal	AGM (AHP)	
	Dr. V Jayan	AGM (EMG)	
	G Hari Narayana Rao	AGM (C&I)	
	Mahesh A	AGM (Civil)	
	Amulya Kumar Behera	AGM (FT & FC)	
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	Piyush Srivastava	AGM (Chemistry)	
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	Ruma De Sharma	AGM (HR)	
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	K Jaganmohan Rao	GM (FGD)	
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	Mathew Eipe Kovoov	GM (Maint)	
Reviewed by	D P Patra	GENERAL MANAGER (O&M)	
Approved for Issue by	Sanjay Kumar Sinha	GROUP GENERAL MANAGER (S)	
Date of Issue			

IMPLEMENTATION OF SAFETY RULES(ELECTRICAL & MECHANICAL)

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IMPLEMENTATION OF SAFETY RULES - ELECTRICAL & MECHANICAL

1.0 Summary

1.1 Superseded Documents: LMI - OPS-SAFT-001 Rev 0

1.2 Scope:

Strict implementation of the safety precautions, procedures, and responsibilities specified in the SR-EM will enable work on or testing of Plant or Apparatus to take place safely. This directive focuses on assurance aspects of the SR-EM and applies to all activities and services carried out in all operational projects of NTPC, that are covered under the SR-EM. For ready reference, the SR-EM previously issued by Corporate-OS, is provided as Annexure-1 of this OD.

In a project where NTPC has outsourced operational responsibility to third-parties, but continues, as the project developer, to have a legal obligation of ensuring a safe working environment, the designated person responsible for such a project shall be accountable for the assurance aspects of this directive to ensure that the philosophy and principles of the SR-EM are implemented in achieving safety from the system by all such third-parties to prevent accidents due to system-derived hazards.

1.3 Review

Review Responsibility : Safety dept.

Review Frequency : 2 yearly basis, or as and when required, whichever is lower

Review Approval : Head of Safety will be reviewing authority and BUH will be approving authority for any revision in this document.

2.0 Introduction:

NTPC Safety Policy recognizes that all accidents are preventable and has as its objective, the provision of a safe working environment and to strive for zero incidents at work. One core objective of the policy, *provision of a safe working environment*, is aimed to be achieved through the clear definition of safety rules governing methods of achieving control over hazards from the interconnected electrical & mechanical system. These have been previously issued by the Corporate-OS and implemented in the form of NTPC Safety Rules (Electrical & Mechanical) Handbook, hereinafter referred to as SR-EM.

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SR-EM are provided to ensure that persons working on plant and apparatus to which these Safety Rules apply are safeguarded from hazards arising from the electro-mechanical system. The SR-EM which are mandatory, are made up of General Provisions and Basic Safety Rules together with sections dealing with Procedures for Safety Documents and Keys, Responsibilities of Persons and Definitions. They are supported by other mandatory and guidance documents like Codes of Practice.

IMPLEMENTATION OF SAFETY RULES - ELECTRICAL & MECHANICAL

This OD covers only the assurance aspects of implementation of aforementioned SR-EM, issued previously. This OD does not contain elements that can vary as per location. Hence no modification or, adaptation for location is expected for this OD.

3.0 Revision Details

In Clause 5.1.1, paragraph 2 is edited as per requirement of procedural compliance Check

4.0 Roles & Accountability

1. Head of the Project/Stations (HOP) is accountable for:

- a. Arrange to have in place a system to ensure the implementation SR-EM, where applicable.
- b. Ensure in place audit processes for such a system at least once a year.
- c. Ensure accountability of Persons appointed under different Roles under the SR-EM.

2. Head of O&M is accountable for:

- a. Appoint of Persons for different roles under the SR-EM.
- b. Ensure audit of the system at least once a year.
- c. Initiate and complete corrective and preventive action to resolve non-compliances including penal action, where required.
- d. Ensure accountability of Persons appointed under different roles.
- e. Identify Persons to carry out compliance audits of the SR-EM implementation quality.

For projects where operations and maintenance functions are outsourced to third parties, the designated Person accountable for the performance of such a project, where ever he/she is located, shall be accountable for the above two roles.

3. Heads of Sections (HOS) are accountable for:

- a. Persons nominated under SR-EM in their department/section are aware of their responsibilities and accountabilities under the SR-EM.
- b. Ensure a system is in place for checking compliance to SR-EM of works in their respective function/department/section.
- c. Maintaining records of compliance checks/audits.
- d. Maintain records of corrective/preventive action, including penal action, taken based on compliance check/audit findings.

4. Persons nominated under different roles shall be accountable for such role as detailed under the SR-EM.

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5. Identified Auditors:

- a. Be thorough with the SR-EM.
- b. Identification and flagging of specific gaps in implementation of SR-EM.

5.0 Requirements

5.1 Compliance checks & audits

Compliance checks & audits of the implementation of SR-EM will be of three types:

1. Procedural Compliance checks.
2. Functional Self-assessments.
3. Compliance checks by senior members of the Project team.
4. Audit by persons outside the project, as part of technical or other audits.

5.1.1 Procedural Compliance Checks

These are checks in the nature of *dip-checks*, made by persons who have been tasked for the purpose, either by appointment or by their functional requirement.

Procedural compliance checks will focus on implementation of the procedural aspects of the SR-EM like completeness of the safety document in all fields, signatures, authorization of the persons who have signed-off, attachment suitability (for e.g., Job Safety Analysis should be specific to the Job Plan under which work is to be carried out), Key-safe keys, isolations completion in the field as per the written document, tool box talk record, PPE, etc.

Procedural checks shall be in the form of random assessments. Auditors shall take random samples from different functions, and where applicable, from different control rooms, to assess the implementation of the procedural aspects. Aspects like failure to implement committed written safety measures shall be flagged as near-miss incidents and processed.

Tool box talk records specific to some tasks shall be checked. Workers' awareness of job specific hazards shall be a check point and lack of workers' awareness of job specific hazards shall be treated as a near-miss and processed accordingly. This is also a check for the effectiveness of the recorded Tool Box talks.

Reports shall be prepared as required by the appointment of the role or that of the function of the auditor and uploaded in to a central repository.

NTPC Limited **Responsibility:**
HOD (EEMG)/(MTP) to include during LMI, Technical & PTW auditing

5.1.2 Functional Self-assessment:

Each HOS who is accountable for implementation of SR-EM in his/her section shall have a schedule of functional self-assessments on compliance to SR-EM. This schedule can also form a part of a larger schedule of self-assessments of the function/department. Self-assessments on compliance to SR-EM will cover all activities under their function's/section's control that are covered by SR-EM.

The assessors shall take random samples from works carried out by their respective section/department/ function, and where applicable, from different control rooms, to assess the quality of the control measures and implementation of the committed control measures. Evidence of non-compliances shall be included in the audit report. Tool box talk records specific to some tasks shall be checked. Workers' awareness of job specific hazards shall be a check point and lack of workers' awareness of job specific hazards shall be treated as a near-miss and processed accordingly.

When a self-assessment has been completed it will be reviewed by the assessor, HOS, HOD and the Head (O&M) to agree on the measures required including penal action, for improving compliance to SR-EM. All such self-assessment records shall be uploaded by the respective function/section as electronic copies that cannot be edited like a secure PDF in to a central repository.

Responsibility:

All HOS, HODs of O&M

5.1.3 Compliance Checks by Seniors of the Project

Head (O&M) who is accountable for implementation of SR-EM shall have a schedule of checks/assessments on compliance to SR-EM. This schedule can also form a part of a larger schedule of self-assessments/audits of the project. Such compliance audit of implementation of SR-EM shall be carried out at least once a quarter by a group of senior persons who independently handle at least one section.

Auditors shall take random samples from different functions, and where applicable from different control rooms, to assess the implementation of the committed control measures. Tool box talk record and associated workers' awareness of job specific hazards shall be a check point and lack of workers' awareness of job specific hazards shall be treated as a near-miss and processed accordingly.

When a compliance assessment has been completed it will be reviewed by the assessors and the Head (O&M) and where suggested by the Head (O&M), by the HOP; to agree on the additional actions to that need to be taken including exemplary penal action to improve the implementation of SR-EM. All such assessment records shall be maintained by the Head (O&M), and uploaded as electronic copies that cannot be edited like a secure PDF in to a central repository.

NTPC Limited

Responsibility:

Head of O&M & his appointed assessors/auditors

5.1.4 Technical and Other Audits

SR-EM implementation compliance audits can be part of a larger scope technical or other audits. This is covered later in this document under [8.0 Audits](#).

Responsibility:

Head of (EEMG)/ (MTP)

5.2 Persons to carry out the compliance checks & audits

The following are the requirements of Persons tasked with compliance checks of SR-EM implementation:

Persons tasked with Procedural Compliance Checks:

- Have rudimentary understanding of essential requirements of SR-EM.

Persons tasked with Functional Self-assessments:

- Preferably, the HOD/HOS should be one of the assessors.
- Possess thorough knowledge of the relevant work, plant, apparatus, methods, equipment, procedures, personnel and workplace.
- Ability to make an assessment of the problem and prioritize any actions.

Senior Persons who carry the SR-EM compliance checks:

- Heads of functions who are thorough with SR-EM.
- Have a good balance of practical skills, theoretical knowledge and actual experience of relevant systems.

Persons from outside the Project who carry the SR-EM implementation audit:

- Possess thorough knowledge of the SR-EM and have a good balance of practical skills, theoretical knowledge and actual experience of implementing SR-EM.

In a project where NTPC has outsourced operational responsibility to third-parties, but continues, as the project developer, to have a legal obligation of ensuring a safe working environment, the designated person responsible for such a project shall be accountable for ensuring persons carrying out periodic audits possess thorough knowledge of the SR-EM along with experience of its practical implementation.

6.0 Records

Any record that is maintained as an electronic copy shall be in a format like secure PDF that cannot be edited.

All HOD/HOS shall maintain: schedule for self-assessments of their respective Function/department; evidence of a system for checking implementation of the SR-EM; evidence of corrective action (including exemplary penal action) taken when safety control measures required as per SR-EM were not implemented. Records of all compliance checks shall be uploaded in a central repository.

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RESPONSIBILITY:

All HOS, HODs of O&M

7.0 Audits

Evaluation of the compliance with this LMI shall be carried out at least once annually. SR-EM implementation compliance audits/checks can be part of a larger scope technical or other audits.

RESPONSIBILITY:

HOD (EEMG)/(MTP)

The audit shall be a combination of verification of: system, records, implementation in the field and interviews.

System: Auditors shall validate the evidence of a system as specified under 6.1.

Records: Audit shall include verification of all the records detailed under 6.1.

Interviews-Nominated Roles: check with at least five employees carrying out nominated roles under the SR-EM like SAP, CP, randomly selected from different functions on their awareness of SR-EM of their respective roles.

HOS of respective section/function shall be accountable for demonstrating compliance to the awareness requirements.

Interviews-Outsourced Persons: Associated outsourced persons' awareness of job specific hazards shall be a check point and lack of awareness of job specific hazards shall be treated as non-compliance and processed accordingly.

HOS of respective section/function shall be accountable for demonstrating the awareness work related hazards of outsourced persons in their function/section.

Field Audit Check points: Auditors shall take a random sample of at least three activities from at least three different functions, and where applicable from different control rooms where safety documents are issued like main plant control room(s), offsite control rooms for coal and ash handling, to assess the implementation of the SR-EM.

Evidence of action initiated, including exemplary penal action, for failure to implement SR-EM shall also be verified and included in the audit report.

All such assessment records shall be maintained by the project-side audit coordinator, and uploaded as electronic copies that cannot be edited like a secure PDF in to a central repository.

In a project where NTPC has outsourced operational responsibility to third-parties, but continues, as the project developer, to have a legal obligation of ensuring a safe working environment, the designated person accountable for such a project shall be accountable for ensuring periodic audits of implementation of the philosophy and principles of the SR-EM are carried out. Records of such audits shall be maintained.

NTPC Limited Responsibility:
All HOS, HODs of O&M

Annexure 1

Safety Rules - Electrical & Mechanical Handbook

The Corporate-OS department had previously issued the NTPC Safety Rules (Electrical & Mechanical) Handbook. The same has been provided here for ready reference and continues from next page.

NTPC Limited



SAFETY RULES HANDBOOK

Electrical and Mechanical

**Corporate Safety Section
National Thermal Power Corporation Ltd.**